

Digital Signal Processing

ASSIGNMENT #1

Q No 1:

Determine the system function of a discrete time system described by the difference equation.

$$y[n] - \frac{1}{3}y[n-1] + \frac{1}{5}y[n-2] = x[n] - 2x[n-1]$$

→ apply z-transform to the equation:

$$Y[z] - \frac{1}{3} \cdot \frac{1}{z} Y[z] + \frac{1}{5} \cdot \frac{1}{z^2} Y[z] = X[z] - \frac{1}{z} X[z]$$

$$Y[z] \left(1 - \frac{1}{3z} + \frac{1}{5z^2} \right) = X[z] \left(1 - \frac{1}{z} \right)$$

$$\frac{Y[z]}{X[z]} = \frac{1 - 1/z}{1 - 1/3z + 1/5z^2} = \frac{z^2 - z}{z^2 - z/3 + 1/5}$$

$$H[z] = \frac{z^2 - z}{z^2 - z/3 + 1/5}$$

←————→

Q No 2:

A causal system has input and output

$$x(n) = \delta(n) + \frac{1}{6}\delta(n-1) - \frac{1}{6}\delta(n-2)$$

$$y(n) = \delta(n) - \frac{2}{3}\delta(n-1)$$

find System function, frequency response and impulse response of system also assess the stability of the system

① System function $H(z)$

taking z-trans of $x[n]$ & $y[n]$

$$X(z) = 1 + \frac{1}{6}z^{-1} - \frac{1}{6}z^{-2}$$

$$Y(z) = 1 - \frac{2}{3}z^{-1}$$

$$H(z) = \frac{1 - \frac{2}{3}z^{-1}}{1 + \frac{1}{6}z^{-1} - \frac{1}{6}z^{-2}}$$

② Impulse response:

$$\frac{1 - \frac{2}{3}z^{-1}}{1 + \frac{1}{6}z^{-1} - \frac{1}{6}z^{-2}} = \frac{A}{\left(1 + \frac{1}{2}z^{-1}\right)} + \frac{B}{\left(1 - \frac{1}{3}z^{-1}\right)}$$

$$1 - \frac{2}{3}z^{-1} = A\left(1 - \frac{1}{3}z^{-1}\right) + B\left(1 + \frac{1}{2}z^{-1}\right)$$

~~$1 - \frac{2}{3}z^{-1}$~~ = put $z^{-1} = -2$

$$1 - \frac{2}{3}(-2) = A\left(1 - \frac{1}{3}(-2)\right) + B\left(1 + \frac{1}{2}(-2)\right)$$

$$\frac{7}{3} = A\left(\frac{5}{3}\right) + 0$$

$$A = \frac{7}{3} \times \frac{3}{5} = \frac{7}{5}$$

put $z^{-1} = 3$

$$1 - \frac{2}{3}(3) = A\left(1 - \frac{1}{3}(3)\right) + B\left(1 + \frac{1}{2}(3)\right)$$

$$-1 = 0 + B\left(\frac{5}{2}\right)$$

$$B = -\frac{2}{5}$$

$$H(z) = \frac{-2/5}{1 - \frac{1}{3}z^{-1}} + \frac{7/5}{1 + \frac{1}{2}z^{-1}}$$

now take The inverse z-trans

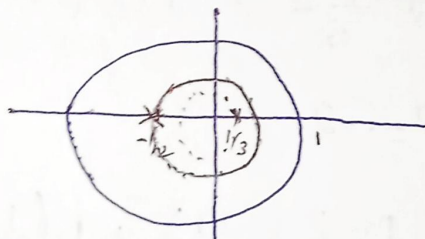
$$h(n) = -\frac{2}{5} \left(\frac{1}{3}\right)^n u(n) + \frac{7}{5} \left(-\frac{1}{2}\right)^n u(n) \quad \text{for } n > 0$$

③ frequency response:

$$H(e^{j\omega}) = \frac{1 - 2/3 e^{-j\omega}}{1 + 1/6 e^{-j\omega} - 1/6 e^{-j\omega}}$$

④ Stability:

we know poles are at $1/3, -1/2$



system is stable as poles are inside the unit circle.

Q NO 3:

A causal system

$$H(z) = \frac{z+2}{2z^2 - 3z + 4}$$

find difference eq & freq. response.

① Difference equation:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z + 2}{2z^2 + 4 - 3z}$$

$$\frac{Y(z)}{X(z)} = \frac{z^{-1} - 2z^{-2}}{2 - 3z^{-1} + 4z^{-2}}$$

$$Y[z] \cdot 2 - 3z^{-1} + 4z^{-2} = X[z] z^{-1} - 2z^{-2}$$

$$2 \cdot Y[z] - 3 \cdot \frac{1}{z} Y[z] + 4 \cdot \frac{1}{z^2} Y[z] = \frac{1}{z} X[z] - 2 \cdot \frac{1}{z^2} X[z]$$

$$2y[n] - 3y[n-1] + 4y[n-2] = x[n-1] - 2x[n-2]$$

② frequency response:

$$H(e^{j\omega}) = \frac{e^{j\omega} + 2}{2e^{j2\omega} - 3e^{j\omega} + 4}$$

Q No 4:

$$x(n) = \left(\frac{1}{3}\right)^n u(n) - \frac{1}{5} \left(\frac{1}{3}\right)^{n-1} u(n-1)$$

$$y(n) = \left(\frac{1}{2}\right)^n u(n)$$

① z-transform

$$X(z) = \frac{1 - \frac{1}{5}z^{-1}}{1 - \frac{1}{3}z^{-1}}, \quad Y(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

② transfer function

$$H(z) = \frac{1 - \frac{1}{5}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{1}{3}z^{-1})}$$

③ Impulse response

$$1 - \frac{1}{3}z = A \left(1 - \frac{1}{5}z \right) + B \left(1 - \frac{1}{2}z \right)$$

$$B = 4/9$$

$$A = 5/9$$

$$H(z) = \frac{5}{9} \left(\frac{1}{1 - \frac{1}{5}z} \right) + \frac{4}{9} \left(\frac{1}{1 - \frac{1}{2}z} \right)$$

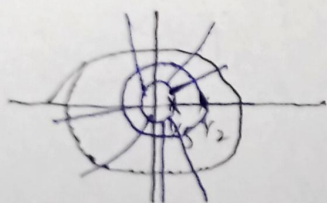
$$H(z) = \frac{5}{9} \left(\frac{1 - \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} \right) + \frac{4}{9} \left(\frac{1 - \frac{1}{5}z^{-1}}{1 - \frac{1}{5}z^{-1}} \right)$$

$$h(n) = \frac{5}{9} \left(\frac{1}{2} \right)^n u(n) + \frac{4}{9} \left(\frac{1}{5} \right)^n u(n)$$

④ freq. response

$$H(e^{j\omega}) = \frac{1 - \frac{1}{3}e^{-j\omega}}{\left(1 - \frac{1}{2}e^{-j\omega} \right) \left(1 - \frac{1}{5}e^{-j\omega} \right)}$$

⑤ stability



system is stable as unit circle is under ROC.